

SEUPD@CLEF: Team RETRIX on Source Retrieval for Scientific Web Claims

Notebook for the CheckThat! Lab at CLEF 2026

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Abstract

We present the results of the RETRIX group for the CheckThat! Lab at CLEF 2026 Task 1, whose goal is to identify the scientific paper implicitly referenced in a social media post. The input consists of short, noisy and usually informal posts that mention scientific findings without providing explicit links or citations. Our pipeline first translates non-English posts into English, then expands them using an embedding model and terms extracted from the top five documents ranked by similarity. After that it is applied a hybrid retrieval system that combines dense FAISS search with sparse lexical matching and ColBERT multi-vector scoring ($s = 0.5 s_{\text{dense}} + 0.15 s_{\text{sparse}} + 0.35 s_{\text{colbert}}$). Finally, we perform neural reranking with a fine-tuned cross-encoder model.

Keywords

Fact Checking, Information Retrieval, Scientific Document Retrieval, Social media fact verification, CLEF, Neural Re-ranking, Cross Encoder

1. Introduction

The rapid spread of scientific claims through social media creates a difficult fact-checking scenario: posts often report, simplify or sensationalize research findings while omitting the original source. In many cases, the cited study is not linked explicitly with a URL or a bibliographic reference but only mentioned indirectly through fragments such as the reported finding, the venue, the institution or the authors. As a consequence, verifying such claims requires first identifying the scientific publication that is being implicitly referenced.

This problem is the focus of Task 1 of the CLEF 2026 CheckThat! Lab, *Source Retrieval for Scientific Web Claims*: given a social media post containing a scientific claim and an implicit reference to a paper, the goal is to retrieve the correct publication from a fixed collection of 10,000 candidate papers written in English.

The task is challenging for several reasons. First, the linguistic gap between social media and scientific writing is substantial: posts are short, informal and often contain paraphrases, slogans, hashtags or incomplete statements whereas scientific papers use more precise and technical language. Second, the reference to the source paper is implicit rather than explicit so the retrieval system must infer the connection from partial semantic evidence rather than from direct citations. Third, relevant clues may appear in different forms, such as mentions of a scientific result, a medical outcome, a research institution or a publication venue, requiring the system to combine lexical and semantic evidence effectively.

An additional challenge is multilinguality: the official query sets include English, German and French posts while the candidate pool remains shared across languages.

Team RETRIX approaches this problem with a four-stage sequential pipeline.

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1. In a preliminary stage we translate to English all the queries written in French and German using the model `utter-project/EuroLLM-9B-Instruct-2512` [1], specifically funded by the EU for understanding and generating text in all European languages. This allows us to handle all queries the same way, independently of the original language and consequently of the semantic and structural differences.
2. Raw tweet-style queries are cleaned of emojis and symbols like "#" and "@" and then enriched via pseudo-relevance feedback: `BAAI/bge-large-en-v1.5` [2] encodes both queries and corpus documents into a shared embedding space, retrieves the top-5 semantically similar papers per query and extracts the most frequent terms from those papers as expansion tokens.
3. The expanded queries are fed into `BAAI/bge-m3` [3] operating in hybrid mode, which combines dense inner-product search via FAISS [4] with sparse lexical matching and ColBERT multi-vector scoring to produce an initial ranked list of candidates.
4. Finally, the top candidates are re-ranked by a neural cross-encoder, in our case `nvidia/llama-nemotron-rerank-1b-v2` [5], fine-tuned on the English training dataset of the CLEF CheckThat! Task 1 2026 and all the queries' datasets of the 2025 CLEF CheckThat! SubTask 4b.

The paper is organized as follows: Section 2 discusses the most relevant research directions related to this task; Section 3 describes in more detail our approach; Section 4 explains our experimental setup; Section 5 discusses our main findings; finally, Section 6 draws some conclusions and outlooks for future work.

2. Related Work

A starting point is lexical retrieval where probabilistic ranking models such as BM25 remain strong baselines thanks to their efficiency, interpretability and robustness in candidate generation [6]. In the setting of scientific claim source retrieval, lexical matching is especially useful when a post mentions recognizable entities or terminology that also appears in the title, abstract, venue or author list of the referenced paper.

Neural retrieval methods have been proposed to mitigate the lexical mismatch between queries and documents by learning dense semantic representations. More broadly, neural information retrieval has shifted attention from exact token overlap toward representation learning and learned matching functions [7]. Within this line of work, dense dual-encoder architectures such as Dense Passage Retrieval (DPR) demonstrated that learned query and document embeddings can be highly effective for first-stage retrieval [8]. This is especially relevant to our task because social media posts often paraphrase scientific findings rather than reuse the exact wording found in the original publication. Building on this direction, BGE-M3 unified dense, sparse, and multi-vector retrieval in a single multilingual model [3], enabling hybrid scoring that combines embedding similarity with lexical token weights, a combination particularly well-suited to cross-lingual settings where queries and documents share few surface tokens.

Multilingual information retrieval presents additional challenges beyond monolingual settings, as systems must bridge language mismatches before relevance can be estimated. Machine translation provides one practical solution, but it can introduce errors and vocabulary drift, especially for technical or domain-specific terminology [9].

Vocabulary mismatch between short, informal queries and longer scientific documents can be further reduced through query expansion. Classical pseudo-relevance feedback methods such as RM3 [10] address this issue by re-weighting query terms using top-ranked documents from an initial retrieval pass. Neural extensions of this paradigm replace or complement lexical evidence with dense representations [11]. Following this general approach, our pipeline incorporates embedding-based query expansion to enrich query representations before scoring against the corpus.

Between initial retrieval and final ranking, many modern systems adopt multi-stage pipelines to balance efficiency and effectiveness [7]. Within these pipelines, late-interaction models such as ColBERT preserve fine-grained token-level matching while remaining substantially more scalable than fully pairwise neural scoring over the whole collection [12]. Similarly, transformer-based re-rankers have

become a common second-stage component for refining an initial candidate set through stronger query-document interactions [13]. Taken together, these approaches motivate retrieval architectures that first generate a manageable set of candidates and then apply more expensive relevance estimation only where it is most useful.

Recent work has argued that reranker fine-tuning benefits from hard negatives because they force the model to learn finer-grained relevance boundaries than easy or random negatives provide [14]. Adapting large language models to retrieval tasks is further facilitated by parameter-efficient fine-tuning with LoRA [15], which has been shown to match or approach full fine-tuning quality for passage reranking while substantially reducing GPU memory requirements [16]. This perspective is also consistent with recent practical guidance for training cross-encoder rerankers with Sentence Transformers [17]. In that setting, fast embedding-based retrieval is used to build a candidate pool, after which the cross-encoder jointly scores query-document pairs and benefits from negatives mined from nearby embedding neighborhoods rather than from trivial mismatches.

3. Methodology

We started from a lexical baseline based on Apache Lucene BM25 and progressively moved to a multi-stage neural pipeline.

The final pipeline (Figure 1) for the proposed information retrieval system, consists of four stages:

- **Query translation:** queries in French or German are first translated into English.
- **Query expansion:** queries are expanded with additional context.
- **Retrieval:** relevant documents are retrieved from the corpus using the expanded query.
- **Reranking:** the retrieved documents are reranked based on their relevance to the original query.

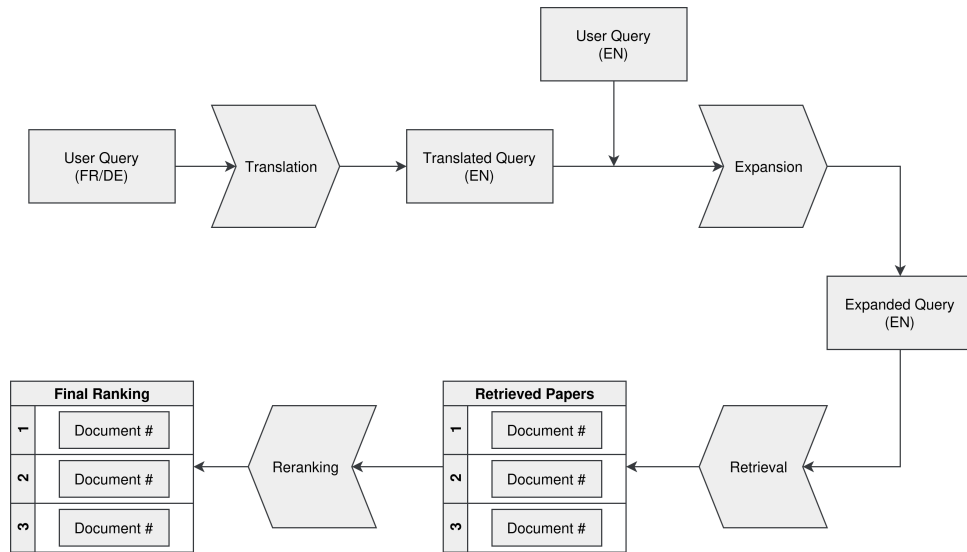


Figure 1: Four-stage pipeline for information retrieval.

3.1. Query translation

In the preliminary stage of our pipeline, non-English queries (French and German) are translated into English. This is done query by query using the model `utter-project/EuroLLM-9B-Instruct-2512` with the following prompt:

LLM Prompt

```
system: You are a professional translator.  
Translate the user's text to English.  
Output ONLY the English translation.  
Do not include the original text, explanations, labels, or any other content.  
  
user: Translate this {French/German} text to English.  
Output only the English translation, nothing else.  
  
{Query original text x}
```

Given x as the original query text, the model predicts probabilities for the next token with [18]:

$$P(y_t = i \mid y_{<t}, x) = \text{softmax}(z_t/T)_i$$

With our model configuration, the prediction becomes greedy, following [1]:

$$y_t = \arg \max_i P(y_t = i \mid y_{<t}, x)$$

The resulting English queries are then fed into the following step of our pipeline.

3.2. Query Expansion

Following the translation stage, each query is processed by a deterministic query-expansion module designed to reduce query–document vocabulary mismatch between short user claims and the terminology used in the scientific corpus. The module does not rely on generative LLM-based paraphrasing. Instead, it combines lexical feature extraction with dense semantic retrieval and neighborhood-based term induction.

- The translated English query is first analyzed with spaCy’s `en_core_web_sm` pipeline, which provides token-level linguistic annotations, including stop-word indicators and coarse-grained part-of-speech labels from the Universal POS tag set. Candidate lexical anchors are then identified by retaining tokens tagged as NOUN or PROPEN while excluding stop words and tokens shorter than three characters. The resulting set is deduplicated to produce a compact lexical signature that captures the principal nominal content of the query.
- The semantic expansion stage operates over a corpus in which each document is represented as the concatenation of its title and abstract. These title–abstract pairs are embedded with BAAI/bge-large-en-v1.5. In accordance with the documented BGE retrieval interface [2], query embeddings are produced by prepending the instruction `{Represent this sentence for searching relevant passages:}` to the translated query, whereas the corpus documents are encoded as retrieval targets. Similarity is then computed on normalized embeddings using the dot product. For unit-length vectors, dot-product scoring is equivalent to cosine similarity while being computationally more efficient.

$$s(q, d) = e_q^\top e_d = \sum_i e_{q,i} e_{d,i}$$

Because embeddings $e = \frac{v}{\|v\|_2}$ are L2-normalized $e_q^\top e_d = \cos(q, d)$ [19].

The five highest-scoring documents are retained as the local semantic neighborhood used for expansion.

In the described implementation, candidate expansion terms are extracted from the retrieved documents after lowercasing, ASCII normalization, mention’s and hashtag’s symbols removal, stop-word filtering and exclusion of short tokens (tokens ≤ 2 characters long are discarded). The candidate pool is aggregated across the retrieved neighborhood and assigned weights derived from the semantic similarity of the source documents so that terms originating from more similar

neighbors contribute more strongly. Terms already present in the query are removed and the fifteen highest-ranking candidates are retained as explicit expansion terms.

$$K = \text{unique}\{\ell(t) : POS(t) \in \{\text{NOUN}, \text{PROPN}\}, \neg stop(t), |t| > 2\}$$

$$w_h = \max(1, \text{round}(5s_h))$$

$$c(t) = \sum_h w_h tf(t, d_h)$$

The final expanded query is constructed by concatenating three components: the cleaned translated query, the noun and proper-noun-based keyword set extracted directly from the query and the expansion terms induced from the retrieved document neighbourhood. This output expanded query is then used by the downstream retrieval pipeline.

3.3. Retrieval

The retrieval stage is designed as a high-recall candidate-generation step between query expansion and neural reranking. Its objective is not to make the final relevance decision but to ensure that the true source paper is included in a sufficiently large candidate set for the subsequent, more computationally expensive cross-encoder stage. For this reason, after experimenting with a Lucene BM25 baseline, the final pipeline adopts **BAAI/bge-m3** in a hybrid retrieval setting.

The retriever searches the fixed paper collection provided by the task organizers. Each paper is represented at document level by concatenating its title and abstract (batch size = 1024) and the retrieved unit is always the paper pubkey. Although the collection includes metadata such as venue and authors, this hybrid retrieval stage uses only titles and abstracts to speed up the inference time.

We chose **BAAI/bge-m3** because it is well suited to this retrieval setting. The model can produce dense vectors, sparse lexical weights and ColBERT-style multi-vector representations from the same encoder.

This allows the retriever to combine broad semantic matching, exact term evidence and fine-grained token-level matching without switching between unrelated encoders.

For the dense representation, the document or query embedding is obtained from the normalized CLS representation [3]:

$$e = \text{norm}(H[0])$$

For the sparse representation, BGE-M3 assigns lexical weights to tokens [20]:

$$w_{qt} = \text{ReLU}(W_{lex}^\top H_q[i])$$

The sparse similarity between a query and a paper is then computed over overlapping terms:

$$s_{lex} = \sum_{t \in q \cap p} w_{qt} w_{pt}$$

Candidate selection is performed in two steps. First, all documents are encoded as dense BGE-M3 vectors and searched with an inner-product FAISS index [4]. The dense FAISS score is computed using inner product [21]:

$$s_{IP}(q, d) = q^\top d$$

Since the dense vectors are normalized, this inner product is equivalent to cosine similarity:

$$q^\top d = \cos(q, d)$$

This dense stage acts as an efficient broad filter over the whole collection. The default setting retrieves a large candidate pool of `candidate_multiplier × top_k`, with `candidate_multiplier = 3` and

`top_k` = 2000, before the final truncation. The number of candidates passed to the hybrid rescoring stage is:

$$K_c = \min(N, \max(K, Km))$$

This deliberately large candidate set favours recall, which is more important than precision at this stage.

Second, the dense candidates are rescored using the sparse and ColBERT-style representations. The sparse component rewards important overlapping terms, while the ColBERT-style component captures local token correspondences that may be lost when each document is represented by a single dense vector [12]. The ColBERT-style late-interaction score is computed as [12]:

$$s_{colbert}(q, d) = \frac{1}{N_q} \sum_{i=1}^{N_q} \max_{1 \leq j \leq N_d} q_i^\top d_j$$

The final configuration was selected through a series of development runs. We first tuned the lexical baseline by varying the Lucene analyzer, stop-word lists, stemming strategy and BM25 searcher variants. These experiments improved the classical baseline but also highlighted its limitations for this task: even with query expansion and reranking, BM25-based candidate generation remained more dependent on surface-level lexical overlap than desired.

We also tested BGE-M3 in a hybrid retrieval setting, using a setup that differed slightly from the final configuration. In those experiments, sparse, dense and ColBERT-style retrieval were performed using dedicated query expansions tailored to each retrieval mode. More specifically, each query was represented through three complementary views: an embedding view for semantic matching, a sparse view for lexical and term-based evidence, and a ColBERT view for fine-grained token-level interaction.

However, the best results in our analysis were obtained with a simpler configuration in which BGE-M3 hybrid retrieval used the same expanded query, described in the previous section, for all three retrieval modes. We then tested several weighting configurations. Among them, the run labelled `bge_m3_hybrid_513` was retained because it provided the strongest recall-oriented trade-off among the retrievers evaluated without reranking. This configuration used weights of 0.5, 0.15 and 0.35 for dense, sparse, and ColBERT scoring, respectively, with a multiplier of 3 and a maximum hybrid input length of 384.

A closely related configuration with a larger ColBERT weight achieved a nearly identical MRR@5 but a lower Recall@100. We therefore preferred the configuration that better matched the role of retrieval as a candidate generator. This choice was further confirmed by downstream experiments: reranking candidates from `bge_m3_hybrid_513` consistently outperformed the corresponding BM25-based reranked runs. In addition, increasing the reranked candidate depth from a few hundred documents to between 1000 and 2500 documents increased the final recall by giving the neural reranker a larger candidate pool.

The final hybrid retrieval score is therefore computed as:

$$s(q, d) = 0.5 s_{dense}(q, d) + 0.15 s_{sparse}(q, d) + 0.35 s_{colbert}(q, d).$$

The largest weight is assigned to the dense score because it provides the main semantic signal. The sparse and ColBERT components add complementary lexical and fine-grained matching evidence. Candidates are sorted by this score in descending order and truncated to the final `top_k`. The retrieval output is a JSON mapping from each query ID to an ordered list of paper pubkeys. Scores are not exported, because the next stage treats retrieval as a candidate generator and recomputes relevance using a cross-encoder over the returned document identifiers.

3.4. Reranking

3.4.1. Cross-encoder scoring

Our strongest single-stage reranker is `nvidia/llama-nemotron-rerank-1b-v2`, a 1B-parameter LLaMA-based model fine-tuned as a sequence classifier. For each query-document pair, we build an

input sequence using the official NVIDIA prompt template:

LLM Prompt

```
question:{query} \n \n passage:{passage}
```

The model computes a scalar logit $z \in \mathbb{R}$ through a linear classification head applied to the final hidden state and the relevance score is obtained by applying a sigmoid:

$$s(q, d) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Documents are then sorted in descending order of $s(q, d)$. In practice, reranking deeper candidate pools (500-2500 documents) consistently produced better final results than shallower pools, confirming the importance of high recall at the retrieval stage to compensate for miss-labeling from weaker previous retrieval-stages. At this stage, queries are evaluated against the corpus of the documents taking into account the title, abstract and author of the documents.

We also evaluated alternative rerankers, including Qwen/Qwen3-Reranker-4B [22] and Alibaba-NLP/gte-reranker-modernbert-base [23, 24] variants.

3.5. Fine-tuning the Reranker

To improve the Nemotron reranker on the domain-specific characteristics of the CheckThat! task, we fine-tuned `nvidia/llama-nemotron-rerank-1b-v2` using Parameter-Efficient Fine-Tuning (PEFT) with Low-Rank Adaptation (LoRA) on data from both the 2025 and 2026 editions of the task.

3.5.1. Training data and hard-negative mining

Training examples are query-document pairs with binary labels: a positive document is the gold source paper associated with the query, while negative documents are hard negatives mined using `sentence-transformers/static-retrieval-mrl-en-v1` [25, 26, 27]. Hard negatives are selected from the range $[5, 50]$ nearest neighbors (by embedding similarity) that satisfy score ≤ 0.85 and lie at least 0.05 below the positive score in cosine similarity (absolute margin) preventing trivially easy or falsely relevant negatives from polluting training that would under-mine the effectiveness of the fine-tuning.

Training data combines two years of the task:

- **2026 data:** split 90% training / 10% development. The development set is drawn exclusively from 2026 data for consistency with the official CT26 Task 1 evaluation protocol. Due to time-constraints we used just the `en_train` dataset but we suppose that also German and French translated queries datasets could improve the quality of the fine-tuning;
- **2025 data:** all available topics (dev, test-gold and train splits) are used entirely for training.

This design ensures that the dev set used to track MRR@5 during training is distribution-aligned with the contest test queries. Each training record contains one query, one positive passage and up to five hard negatives.

3.5.2. LoRA architecture

To fine-tune the 1B-parameter base model with limited GPU memory, we apply LoRA [15] with rank $r = 16$ and scaling factor $\alpha = 32$. Adapters are inserted into all attention and feed-forward projection layers: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj` and `down_proj`. The classification head (score) is included in `modules_to_save` and is fully updated during training. The task type is `SEQ_CLS`, matching the sequence-classification objective of the reranker.

3.5.3. Training objective

The model produces a single scalar logit z per input pair. Training minimizes binary cross-entropy with logits over all positive ($y = 1$) and negative ($y = 0$) pairs in each batch:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \left[y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right]$$

where B is the effective batch size and σ is the sigmoid function. Gradient accumulation over 4 steps is used to simulate a larger effective batch size of 32 without additional memory overhead.

3.5.4. Optimization

The model is trained with AdamW, a learning rate of 2×10^{-5} , a linear warmup over 10% of the total training steps and gradient clipping at norm 1.0. Training runs for 3 epochs with bfloat16 mixed precision and gradient checkpointing enabled to reduce VRAM usage. The checkpoint with the best MRR@5 on the 2026 development set is retained.

3.5.5. Inference

At inference time, the LoRA adapter weights are merged into the base model via `merge_and_unload()` before scoring, so no PEFT overhead is incurred during evaluation. The scoring procedure and prompt format are identical to those of the base reranker (Section 3.4) and the fine-tuned model is used as a drop-in replacement in the reranking stage.

3.6. Evaluation

We evaluate systems on the CheckThat! Task 1 query sets by matching retrieved pubkeys against the gold pubkey for each query. The official evaluation metric for the task is MRR@5. We also tracked Recall@100 to monitor candidate-generation quality and to verify that reranking gains do not come from overly aggressive filtering.

4. Experimental Setup

All experiments were conducted on our own machines, using a computer with an AMD Ryzen 9 5900X, 32 GB of RAM and an NVIDIA RTX 3090 GPU (10,496 CUDA cores, 1.70 GHz boost clock, 24 GB of GDDR6X memory, 35.58 TFlops) for resource-intensive workloads and our laptops for standard workloads and program refinement. We used four laptops: the two most powerful are a MacBook Air M3 and a machine with no dedicated GPU, 16 GB of RAM and an Intel Core i7-13700H. For reproducibility, all environment settings have been documented and are available in the Bitbucket repository at <https://bitbucket.org/upd-dei-stud-prj/seupd2526-retrix/src/master/> or can be reproduced by following the instructions in this paper.

4.1. Dataset

The dataset consisted of 10,000 scientific publications that had to be paired with tweet-like posts written in three languages: German, French, and English. There are no explicit references such as URLs in any query. Instead, each tweet contains an implicit reference to a scientific paper, for example through mentions of authors or keywords related to the paper. The dataset files are all in JSON format. The scientific papers dataset contains the columns pubkey (unique identifier), title, abstract, venue/location and authors. The query datasets contain the columns index, pubkey (ID of the positive scientific paper that has to be matched with the query) and text.

Table 1
Dataset statistics by language

Split	Language	File	Queries		Length (tokens)			
			#	%	Min	Median	Max	Mean
Train	EN	en_train.json	14977	77.8	2	33	57	31.5
	FR	fr_train.json	2807	14.6	2	33	57	31.5
	DE	de_train.json	1460	7.6	2	33	57	31.5
Total			19244	100	2	33	57	31.5
Dev	EN	en_dev.json	3905	78.2	5	33	60	31.7
	FR	fr_dev.json	702	14.1	5	33	60	31.7
	DE	de_dev.json	386	7.7	5	33	60	31.7
Total			4993	100	5	33	60	31.7
Test	EN	final_en_test.json	6076	74.4	5	33	58	31.9
	FR	final_fr_test.json	1220	14.9	5	33	58	31.9
	DE	final_de_test.json	876	10.7	5	33	58	31.9
Total			8172	100	5	33	58	31.9
Overall Total			32409		2	33	60	31.7

Figure 2 compares lengths distribution of tokens across train, dev and test datasets. All three datasets share similar distributions, with the same median value and similar length value peaks.

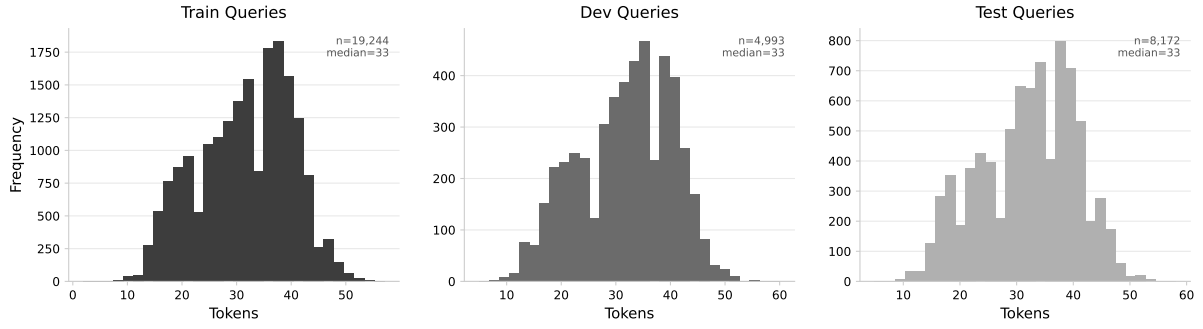


Figure 2: Train, dev and test datasets’ distribution of query token length.

5. Results and Discussion

5.1. Main Results

Table 2 summarizes representative configurations from the run-analysis sheets performed. The table is not meant to list every run we have made; instead, it selects runs that cover the main experimental families: analyzer tuning, hybrid retrieval, single-stage neural reranking, and alternative rerankers. The comparison uses the English run-analysis sheet with 14,977 evaluated queries from the `train_set`, and all metric values are rounded to three decimals.

In the analyzer rows, `MyEnglishAnalyzer_V2` denotes the Lucene analyzer using the RankSNL large stopwords list, ASCII folding, lowercasing, English possessive removal, and KStem. Searcher V2 denotes the BM25 searcher variant that adds Boolean OR clauses.

Table 2

Representative English run-analysis results performed on `en_train_set`.

Family	Run label	Key configuration	MRR@5	Recall@100
Analyzer tuning	Analyzer V2	BM25 retrieval with <code>MyEnglishAnalyzer_V2</code> , Searcher V2, and BGE-V1 query expansion	0.503	0.796
Analyzer tuning	Analyzer V1.12	BM25 retrieval with KStem and an EBSCO-host/MEDLINE/CINAHL stoplist variant	0.450	0.783
Retrieval only	Hybrid A	BGE-M3 hybrid retrieval with dense, sparse, and ColBERT weights of 0.35, 0.15, and 0.50	0.557	0.837
Retrieval only	Hybrid B	BGE-M3 hybrid retrieval with dense, sparse, and ColBERT weights of 0.50, 0.15, and 0.35	0.557	0.839
Best model reranking	Nemotron top-100	<code>bge_m3_hybrid_513</code> candidates reranked with <code>nvidia/llama-nemotron-rerank-1b-v2</code>	0.691	0.839
Best model reranking	Nemotron top-1000	<code>bge_m3_hybrid_513</code> candidates reranked with <code>nvidia/llama-nemotron-rerank-1b-v2</code>	0.702	0.911
Alternative rerankers	Qwen3-4B top-100	<code>bge_m3_hybrid_513</code> candidates reranked with <code>Qwen3-reranker-4B</code>	0.662	0.839
Alternative rerankers	GTE-ModernBERT top-100	<code>bge_m3_hybrid_513</code> candidates reranked with <code>Alibaba-NLP/gte-reranker-modernbert-base</code>	0.652	0.852

The results show a consistent progression from lexical retrieval toward neural ranking. The best retrieval-only configuration reaches an MRR@5 of 0.557, while the strongest reranked run reported in Table 2 reaches 0.702. The largest observed step comes from adding `nvidia/llama-nemotron-rerank-1b-v2`: the top-1000 Nemotron run reaches an MRR@5 of 0.702 and Recall@100 of 0.911, compared with 0.557 and 0.839 for the retained hybrid retriever. We do not report confidence intervals, paired significance tests, or effect sizes for these aggregate comparisons, so they should be read as observed metric differences only. Due to limited computing resources and time constraints, we were not able to run the same setting on the full English training dataset; it would have required more than 40 hours of computation on our RTX 3090 for the top-k = 1000 reranking.

For the runs performed on the dev_set, we provide a separate multilingual check rather than a single aggregate score. On the English development split, the non-fine-tuned Nemotron baseline reaches 0.675 MRR@5 and 0.835 Recall@100 when reranking the top 100 candidates; it reaches 0.692 MRR@5 and 0.909 Recall@100 when reranking the top 1000 candidates. The fine-tuned Nemotron LoRA run on the same English development split reaches 0.716 MRR@5 and 0.913 Recall@100. The corresponding fine-tuned development runs reach 0.76 MRR@5 and 0.929 Recall@100 for French and 0.67 MRR@5 and 0.889 Recall@100 for German with the top-2000 candidate setting.

5.2. Evaluation Evidence and Metrics

We report three complementary forms of evidence. First, we use the TRAIN English run-analysis split to compare retrieval and reranking configurations over 14,977 queries. This is the largest labeled split available to us and is therefore the most reliable source for per-query statistical analysis. Second, we validate the final fine-tuned reranker on the multilingual DEV splits: English, French, and German. Third, we evaluate the final official TEST submission; this lets us report effectiveness metrics for the final multilingual runs.

The primary metric is MRR@5, following the task objective. We also report Recall@100 to measure whether the correct paper remains in the candidate set, and MAP and nDCG@10 to describe the quality of the ranking beyond the first reciprocal-rank hit. Since each query has a single gold publication in the available labeled files, per-query MRR@5 is zero unless the gold paper appears in the first five positions, and is $1/r$ when the first relevant paper appears at rank $r \leq 5$.

5.3. Run Comparison on the TRAIN English Split

In this section, we further analyze the effects of varying the parameters and pool depth of the main pipeline components. All modified parameters are reported in Table 3. The hybrid runs differ in the linear weights assigned to dense, sparse, and ColBERT-style BGE-M3 signals. The reranking runs use the retained hybrid candidate generator H513 and rerank increasingly deep candidate pools with `nvidia/llama-nemotron-rerank-1b-v2`. The Δ column is computed against H513, because this is the hybrid configuration used by the submitted reranking pipeline.

Table 3

Run comparison on the TRAIN English split with 14,977 queries. Values are computed from ranked outputs and the corresponding gold labels.

Run	Family	Configuration	MRR@5	Recall@100	MAP	nDCG@10	Δ MRR@5
H424	Hybrid	BGE-M3 weights 0.40/0.20/0.40	0.5067	0.8277	0.5201	0.5531	-0.0500
H514	Hybrid	BGE-M3 weights 0.50/0.10/0.40	0.5141	0.8327	0.5273	0.5595	-0.0426
H414	Hybrid	BGE-M3 weights 0.45/0.10/0.45	0.5171	0.8327	0.5302	0.5621	-0.0396
H513	Hybrid	BGE-M3 weights 0.50/0.15/0.35	0.5567	0.8392	0.5688	0.5956	0.0000
H315	Hybrid	BGE-M3 weights 0.35/0.15/0.50	0.5572	0.8365	0.5692	0.5957	+0.0005
N100	Reranking	H513 candidates reranked with Nemotron, top-100 retained	0.6847	0.8392	0.6909	0.7146	+0.1280
N500	Reranking	H513 candidates reranked with Nemotron, top-500 retained	0.7010	0.8986	0.7096	0.7342	+0.1443
N1000	Reranking	H513 candidates reranked with Nemotron, top-1000 retained	0.7018	0.9109	0.7111	0.7357	+0.1451

The weakest configuration among the reported runs is H424, with MRR@5 of 0.5067. Increasing the dense component while keeping a moderate sparse contribution is beneficial: H513 improves MRR@5 by 0.0500 over H424 and also reaches the best Recall@100 among the hybrid-only runs. H315 has a marginally higher MRR@5 than H513, but its Recall@100 is lower. For this reason, H513 is the more appropriate candidate generator for the full pipeline, where the goal of retrieval is to expose as many plausible documents as possible to the reranker.

The largest improvement is introduced by neural reranking. Moving from H513 to N1000 increases MRR@5 from 0.5567 to 0.7018, an absolute gain of 0.1451 and a relative improvement of 26.1%. The gain is not only an early-rank effect: Recall@100 also rises from 0.8392 to 0.9109 when reranking a deeper

candidate pool. The difference between N500 and N1000 is much smaller in MRR@5, only 0.0008, but N1000 keeps a higher Recall@100. This indicates diminishing returns in early precision after roughly 500 reranked candidates, while deeper reranking still helps recover relevant documents in the tail.

Paired Wilcoxon signed-rank tests with Holm correction on per-query MRR@5 corroborate these observations. Specifically: H315 vs H513 is not significant (Holm-adjusted $p = 1.0$). By contrast, H513 outperforms H424 with a highly significant difference (Holm-adjusted $p \approx 8.58 \times 10^{-116}$). The improvement from H513 to the Nemotron top-1000 reranker is both large and highly significant (mean $\Delta = 0.1451$; Holm-adjusted $p = 0.0$). The difference between N500 and N1000 is statistically detectable after Holm correction (Holm-adjusted $p \approx 0.04974$) but its practical magnitude on MRR@5 is negligible ($\Delta \approx 0.0008$), which supports the "diminishing returns" interpretation. For Recall@100 the H513 vs N1000 gain is also significant (Holm-adjusted $p \approx 3.46 \times 10^{-195}$).

5.4. Multilingual DEV Split Validation

Table 4 reports the final fine-tuned Nemotron LoRA reranker on the multilingual development splits. These data are closer to the official multilingual TEST setting than the English-only run-analysis split, because they include translated French and German queries. The confidence intervals are normal-approximation 95% intervals over per-query MRR@5 values; they are descriptive rather than official task intervals.

To be noted that we have not performed runs with fine-tuned Nemotron LoRA on the English run-analysis split, so the DEV results cannot be directly compared with the TRAIN runs in Table 3. The DEV results are meant to validate the final pipeline on a multilingual setting, not to compare with the English-only runs. The DEV splits are also smaller than the TRAIN split, which leads to wider confidence intervals and less stable estimates.

Table 4

Fine-tuned Nemotron LoRA validation on labeled development data. The final two columns report rank-bucket counts for strong successes and unresolved failures.

Split	Queries	MRR@5	95% CI	Recall@100	MAP	nDCG@10	Rank 1 / Missing
EN DEV	3905	0.7160	[0.7030, 0.7290]	0.9129	0.7243	0.7483	2574 / 241
FR DEV	702	0.7599	[0.7310, 0.7888]	0.9288	0.7672	0.7894	496 / 40
DE DEV	386	0.6736	[0.6305, 0.7168]	0.8860	0.6820	0.7080	237 / 29

The development results suggest that the final pipeline generalizes reasonably across languages after translation, but not uniformly. French is the strongest development split, with MRR@5 of 0.7599 and Recall@100 of 0.9288. German is the weakest, with MRR@5 of 0.6736 and the widest confidence interval because it has only 386 labeled queries. This lower German score is a warning for the official TEST set: the German portion is smaller than English but still accounts for 876 of the 8172 test queries, and translation or morphology errors can have a visible effect on aggregate performance.

5.5. Official TEST Set Submission

The official TEST input used for submission consists of 8172 queries: 6076 in English, 1220 in French, and 876 in German. The submitted systems use the same four-stage pipeline described in Section 3: translate non-English queries into English, expand the query, retrieve candidates with BGE-M3 hybrid retrieval, and rerank with the fine-tuned Nemotron LoRA cross-encoder.

The results obtained from the official TEST set confirm that the final system is effective rather than merely well-formed; French is the strongest TEST split, followed by English, while German remains the weakest. This mirrors the development pattern in Table 4;

On the TEST set, the reranking is computed from the top-2000 candidates retrieved by the BGE-M3 hybrid retriever, which is a deeper pool than the top-1000 used for the TRAIN runs. This is because the DEV results suggested that deeper reranking can still improve Recall@100, and we wanted to maximize

Table 5

Official TEST set results metrics. The aggregate row is a micro-average over all queries.

Language	MRR@5	Recall@100	MAP	nDCG@10
EN	0.7117	0.9235	0.7202	0.7469
FR	0.7400	0.9328	0.7479	0.7689
DE	0.6806	0.9155	0.6907	0.7213
Aggregate	0.7126	0.9240	0.7212	0.7474

recall for the final submission. The improvement from top-1000 to top-2000 candidates on the DEV set was modest but positive, so we opted for the deeper pool in the TEST submission to give the reranker more chances to find relevant documents. The final TEST results are consistent with the DEV results, which suggests that the pipeline generalizes reasonably well to unseen queries and that the translation step does not introduce excessive noise.

5.6. Statistical Analysis

We separate descriptive evidence from hypothesis testing. The descriptive part characterizes the distribution of per-query MRR@5 scores, while the paired tests compare systems on the same queries. This is important because aggregate metrics alone hide whether an improvement is broad across many queries or concentrated in a smaller number of large corrections.

5.6.1. Descriptive statistics

Table 6 reports per-query MRR@5 statistics for the retained hybrid run and the Nemotron reranking depths. The reranker changes the distribution, not only the mean: the median rises from 0.5 for H513 to 1.0 for all Nemotron runs, and the first quartile rises from 0.0 to 0.25 after reranking. This means that at least 75% of queries receive a non-zero MRR@5 score after reranking, whereas the hybrid-only system still has a first quartile of zero.

Table 6

Descriptive statistics of per-query MRR@5 on the TRAIN English split. Confidence intervals use the normal approximation over queries.

Run	Mean	Median	Std. dev.	Variance	Min	Q1	Q3	95% CI
H513	0.5567	0.5	0.4614	0.2129	0.0	0.00	1.00	[0.5493, 0.5641]
N100	0.6847	1.0	0.4303	0.1852	0.0	0.25	1.00	[0.6778, 0.6916]
N500	0.7010	1.0	0.4229	0.1788	0.0	0.25	1.00	[0.6943, 0.7078]
N1000	0.7018	1.0	0.4223	0.1784	0.0	0.25	1.00	[0.6950, 0.7086]

Figure 3 provides a direct visual comparison across the English statistical-analysis runs; on the training split the confidence intervals of H315 and H513 nearly coincide, confirming that the two retained hybrid settings are essentially tied on MRR@5 even though they differ slightly in Recall@100. The same figure also shows that the gain from reranking is not marginal: the Nemotron family shifts the mean from roughly 0.56 to roughly 0.69–0.70. On the development split, the fine-tuned Nemotron LoRA model reaches 0.716 MRR@5 with a 95% interval of [0.703, 0.729], clearly above the GTE/ModernBERT baseline and modestly above the untuned Nemotron runs.

5.6.2. Rank-bucket and error analysis

Table 7 reports the rank position of the gold document for the retained hybrid and reranked runs. The analysis separates three qualitatively different error types. A gold paper is counted as *missing* only

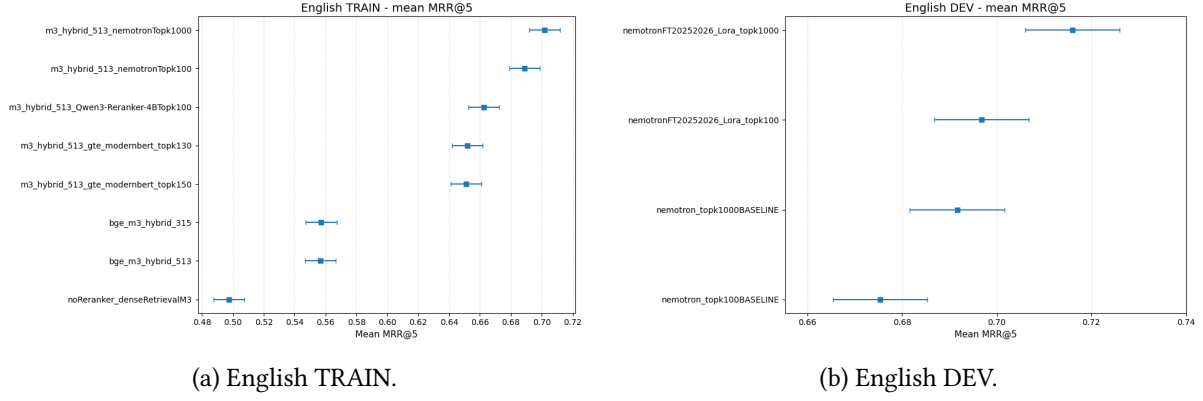


Figure 3: Mean MRR@5 with normal-approximation 95% confidence intervals for the English TRAIN and DEV statistical-analysis runs.

when its gold pubkey is absent from every document available in the ranked file for that query. Thus, for runs with retained deep candidate lists, missing cases indicate that the reranker cannot recover the document from its input pool. For runs available only as submitted top-100 lists, missing means absent from the available top-100 file, not necessarily absent from the original upstream retrieval pool. A gold paper appearing after rank 100 has been retrieved by the retained reranking pool but lies outside the Recall@100 cutoff. A gold paper appearing between ranks 6 and 100 is present in the candidate evidence but is ranked too low to satisfy the top-5 objective.

Table 7

Rank buckets for the retained hybrid and reranked English runs. Percentages are computed within each split: 14,977 labeled English queries, 3,905 English DEV queries, and 6,076 English TEST queries.

Split	Family	Queries	Rank 1	Ranks 2–5	Ranks 6–10	Ranks 11–100	Rank > 100	Missing
EN TRAIN	Hybrid BGE-M3	14977	7459 (49.8%)	2280 (15.2%)	715 (4.8%)	2114 (14.1%)	0 (0.0%)	2409 (16.1%)
EN TRAIN	Nemotron Baseline	14977	9697 (64.7%)	2057 (13.7%)	598 (4.0%)	1291 (8.6%)	405 (2.7%)	929 (6.2%)
EN DEV	Nemotron LoRA fine-tuned	3905	2574 (65.9%)	538 (13.8%)	144 (3.7%)	309 (7.9%)	99 (2.5%)	241 (6.2%)
EN TEST	Nemotron LoRA fine-tuned	6076	3950 (65.0%)	926 (15.2%)	229 (3.8%)	506 (8.3%)	0 (0.0%)	465 (7.7%)

The labeled English comparison shows that the main effect of reranking is to move correct papers to the very top of the list. The best reranked run on the TRAIN split places the gold paper at rank 1 for 9697 queries, compared with 7459 for the retained hybrid single-stage run. It also reduces missing cases from 2409 to 929. The remaining errors are informative: for Nemotron Baseline, 1291 queries have the gold paper in ranks 11–100, indicating ranking failures rather than retrieval failures, whereas 929 queries remain missing from the available reranking pool and therefore cannot be recovered by the reranker.

The final English DEV and TEST runs follow the same pattern. Nemotron LoRA fine-tuned places the gold paper at rank 1 for 2574 of 3905 queries, and does so also for the TEST set with 3950 of 6076 queries. Their top-5 coverage is similar, with 3112 DEV queries and 4876 TEST queries resolved within the first five ranks. The DEV file contains 1000 ranked documents for each query, so the 99 cases in Rank > 100 correspond to gold papers found by the retained reranking pool but not placed within the first 100 positions. The TEST file used to compute the metrics contains only the submitted top-100 documents per query; consequently, Rank > 100 is zero by construction, and the 465 missing TEST cases should be interpreted as missing from the available submitted run rather than as evidence that the original upstream retrieval pool never retrieved the gold paper.

Overall, the bucket analysis indicates that reranking substantially improves early precision while leaving two residual failure modes. The first consists of ranking failures, where the gold paper is present but remains below the top-5 cutoff. The second consists of missing cases, which are more severe because they cannot be repaired by reranking alone. These missing cases are likely associated with severe vocabulary mismatch, translation drift, or claims whose wording emphasizes entities and contextual

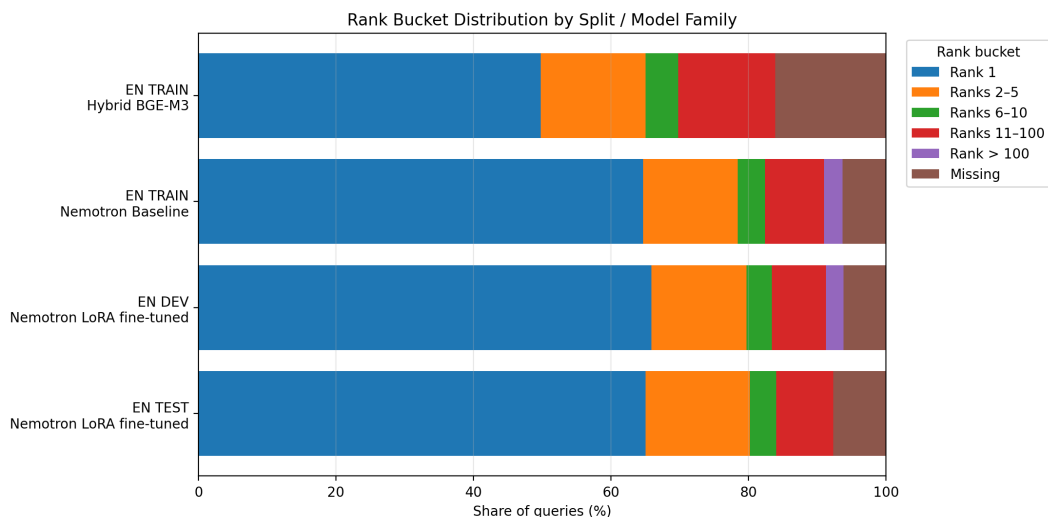


Figure 4: Visual representation of the rank bucket distribution for the retained hybrid and Nemotron runs on the EN TRAIN, DEV and TEST splits.

cues that are weakly represented in titles and abstracts.

In Figure 4, a visual representation of the distribution is shown.

5.6.3. Error patterns and failure analysis

We manually inspected missing cases as a qualitative diagnostic. The following patterns are not mutually exclusive and should be interpreted as plausible failure modes rather than a measured taxonomy:

1. **Claim-to-metadata specificity mismatch:** Some claims mention specific findings, medical outcomes, or statistical results that are not present in the indexed title and abstract metadata. These cases are difficult for the first-stage retriever because the query emphasizes details unavailable in the indexed document representation.
2. **Translation artifacts in non-English queries:** In French and German runs, some translated queries contain domain-term shifts, awkward literal translations, or untranslated technical expressions. These artifacts can reduce lexical and semantic alignment with the English title–abstract corpus.
3. **Implicit references with little lexical overlap:** Some claims refer to a paper indirectly, using phrases such as “this study” or “these papers” without naming concrete findings, authors, institutions, or venues. In these cases, the social-media wording provides too little evidence for reliable candidate retrieval.

5.6.4. Performance by query characteristics

Figure 5 reports a query-level characterization of the Nemotron Baseline run on the English TRAIN split. Contrary to our initial intuition, token length alone is not positively associated with higher MRR@5 in this run: short queries with at most 20 tokens reach 0.760 MRR@5, medium-length queries reach 0.713, and long queries with at least 35 tokens reach 0.672. This suggests that longer tweet-like claims often add conversational or contextual material that does not necessarily help identify the target paper.

The named-entity cue analysis is more consistent with the retrieval intuition. Using a lightweight heuristic based on author-like capitalization patterns, acronyms, and institution or venue keywords, entity-rich queries reach 0.733 MRR@5, while entity-sparse queries reach 0.664. These observations remain exploratory and should be interpreted as heuristic patterns rather than formal statistical findings, but they support the value of hybrid retrieval that combines lexical anchors with semantic matching.

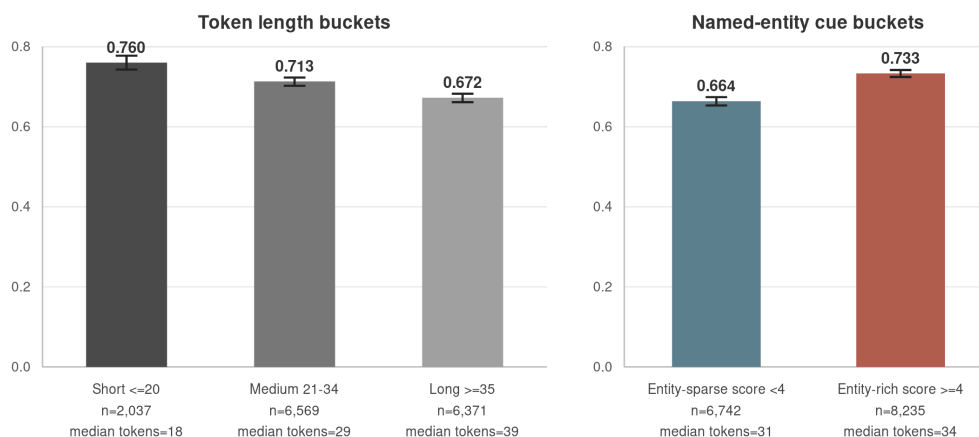


Figure 5: Mean per-query MRR@5 for the Nemotron Baseline English TRAIN run grouped by query length and by a lightweight named-entity cue score. Error bars show normal-approximation 95% confidence intervals over queries.

6. Conclusions and Future Work

The results of the described system support three main conclusions.

First, analyzer and BM25 tuning provide a useful lexical baseline, especially thanks to its low computational footprint, with the best analyzer-focused run reaching 0.503 MRR@5 and 0.796 Recall@100 in the English run analysis.

Second, moving to BGE-M3 hybrid retrieval produces a stronger candidate generator: the retained configuration, `bge_m3_hybrid_513`, reaches 0.557 MRR@5 and 0.839 Recall@100.

Third, reranking the hybrid candidates with `nvidia/llama-nemotron-rerank-1b-v2` gives the largest observed improvement among the reported single-stage runs, reaching 0.702 MRR@5 and 0.911 Recall@100 when reranking the top 1000 candidates. Alternative rerankers such as `Qwen3-reranker-4B` and `Alibaba-NLP/gte-reranker-modernbert-base` also improve results, but remain below the Nemotron runs in the reported MRR@5 values.

The development results show that fine-tuning the Nemotron reranker is promising. Fine-tuning improves performance by a small but appreciable amount (an average gain of 0.03 MRR@5 points in our experiments).

6.1. Future Work

Several directions emerge for future improvement. First, analyzing and spending more time for a better fine-tuning of the reranker and all other components of the system could bring better results. A lot of techniques like data augmentation can be experimented and there is room for improvement.

A second direction concerns the queries that remain entirely missing after reranking. These cases cannot be recovered by any reranker because the correct publication is absent from the candidate set. Manual inspection of a sample suggests three dominant failure patterns: claims referencing quantitative findings not mentioned in the abstract, implicit references that name no concrete entities and translation drift that changes the intended meaning of domain-specific terms. Addressing the first two patterns would probably require to manage content beyond title, abstract and authors but also other metadata. Addressing the translation failure pattern points toward a dedicated post-editing step or a domain-adapted translation model fine-tuned on biomedical text.

Finally, the German dataset achieves MRR@5 of 0.68 in the evaluation results: a wide gap relative to English and French. Controlled ablations that isolate translation errors from retrieval errors would

clarify how much of this gap is attributable to the translation stage versus vocabulary mismatch in the retrieval stage itself.

7. Declaration on Generative AI

During the writing of this paper the authors used Grammarly and DeepL for grammar and spelling checks and Claude Sonnet 4.6 for writing-style improvement, citation management, and formatting assistance. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the publication content.

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